Tarea S3.01. Manipulación de tablas

Nivel 1

- Ejercicio 1

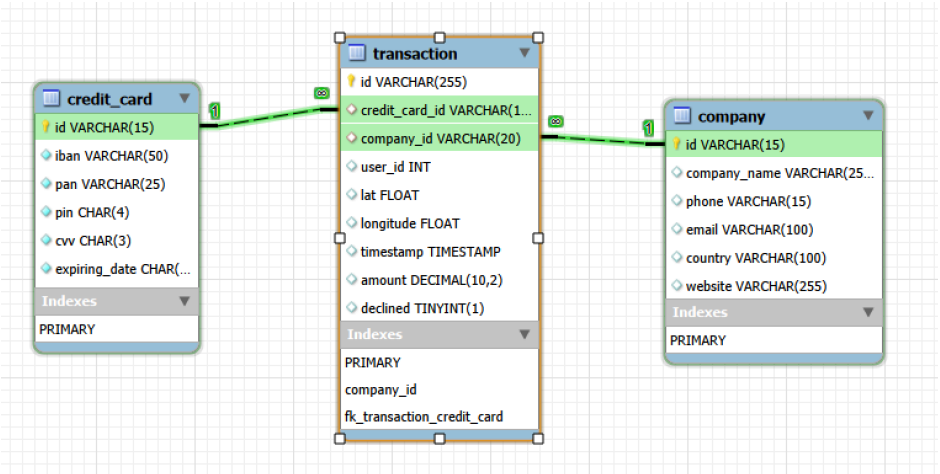
Tu tarea es diseñar y crear una tabla llamada "credit_card" que almacene detalles cruciales sobre las tarjetas de crédito. La nueva tabla tiene que ser capaz de identificar de manera única cada tarjeta y establecer una relación adecuada con las otras dos tablas("transaction" y "company"). Después de crear la tabla será necesario que ingreses la información del documento denominado "datos_introducir_sprint3_credit". Recuerda mostrar el diagrama y realizar una breve descripción de este.

```
CREATE TABLE IF NOT EXISTS credit_card (
    id VARCHAR(15) PRIMARY KEY,
    iban VARCHAR(50) NOT NULL,
    pan VARCHAR(25) NOT NULL,
    pin CHAR(4) NOT NULL,
    cvv CHAR(3) NOT NULL,
    expiring_date CHAR(8) NOT NULL
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105113	10:05:26	SELECT * FROM transactions.credit_card LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.0
105114	10:16:43	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(15) PRIMARY KEY, iban VARCHA...	0 row(s) affected	0.062 sec



En este diagrama, tenemos la tabla de hechos: **transaction** y las de dimensiones: **company** y **credit_card**

company y **credit_card** tienen una relación 1 a N con **transaction**.

Los campos de la tabla **credit_card**

Campo	Tipo	Propiedades	Descripción
id	Varchar(15)	Primary Key	Mismo tipo de dato que el campo **credit_card_id** de la tabla **transaction**, con el que está vinculado
iban	Varchar(50)	NOT NULL	Puede tener diferentes longitudes según el país, normalmente hasta 35
pan	Varchar(25)	NOT NULL	Puede tener diferentes longitudes según la tarjeta, normalmente entre 12-19
pin	Var(4)	NOT NULL	Contiene 4 caracteres numéricos, puede tener 0's a la izquierda
cvv	Var(3)	NOT NULL	Contiene 3 caracteres numéricos, puede tener 0's a la izquierda
expiring_date	Var(8)	NOT NULL	Su formato es 'dd/mm/yy' de longitud fija.

```
66
67 •  ALTER TABLE transaction
68    ADD CONSTRAINT fk_transaction_credit_card
69    FOREIGN KEY (credit_card_id)
70    REFERENCES credit_card( id );
```

Output

Action Output

#	Time	Action	Message	Duration / Fet
105111	09:35:16	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX9156705...	1 row(s) affected	0.015 sec
105112	09:48:18	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_credit_card FOREIGN KEY (credit_c...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.969 sec
105113	10:05:26	SELECT * FROM transactions.credit_card LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0

Añado la foreign key a la tabla **transaction** vinculando el campo **credit_card_id** con la PRIMARY KEY **id** de la tabla **credit_card**.

- Ejercicio 2

El departamento de Recursos Humanos ha identificado un error en el número de cuenta asociado a la tarjeta de crédito con ID CcU-2938. La información que tiene que mostrarse para este registro es: TR323456312213576817699999. Recuerda mostrar que el cambio se realizó.

```
29
30 •  SELECT * FROM credit_card WHERE id = 'CcU-2938';
31 •  UPDATE credit_card
32    SET iban = 'TR323456312213576817699999'
33    WHERE id = 'CcU-2938';
34 •  SELECT * FROM credit_card WHERE id = 'CcU-2938';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	iban	pan	pin	cvv	expiring_date
CcU-2938	TR323456312213576817699999	5424465566813633	3257	984	10/30/22
NULL	NULL	NULL	NULL	NULL	NULL

credit_card 2 | credit_card 3 | Apply

Output

Action Output

#	Time	Action	Message	Duration / F
5026	12:29:10	SELECT * FROM credit_card WHERE id = 'CcU-2938' LIMIT 0, 1000	1 row(s) returned	0.000 sec /
5027	12:29:10	UPDATE credit_card SET iban = 'TR323456312213576817699999' WHERE id = 'CcU-2938'	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
5028	12:29:10	SELECT * FROM credit_card WHERE id = 'CcU-2938' LIMIT 0, 1000	1 row(s) returned	0.000 sec /

- Ejercicio 3

En la tabla **transaction** ingresa una nueva transacción con la siguiente información:

Id	108B1D1D-5B23-A76C-55EF-C568E49A99DD
credit_card_id	CcU-9999
company_id	b-9999
user_id	9999
lat	829.999
longitude	-117.999
amount	111.11
declined	0

```
INSERT INTO transaction
( id, credit_card_id, company_id, user_id, lat, longitude, amount, declined )
VALUES( '108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, 111.11, 0 );

INSERT INTO company (id) VALUES ('b-9999');
INSERT INTO credit_card VALUES ('CcU-9999','for testing','for testing','test','tes','xx/xx/xx');

INSERT INTO transaction
( id, credit_card_id, company_id, user_id, lat, longitude, amount, declined )
VALUES( '108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, 111.11, 0 );
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	12:22:54	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) V...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`testing`.`transaction`, ...	0.032 sec
2	12:22:59	INSERT INTO company (id) VALUES ('b-9999')	1 row(s) affected	0.015 sec
3	12:22:59	INSERT INTO credit_card VALUES ('CcU-9999','for testing','for testing','test','tes','xx/xx/xx')	1 row(s) affected	0.000 sec
4	12:23:04	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) V...	1 row(s) affected	0.016 sec

El primer intento de inserción da error por no cumplir con las restricciones de las claves foraneas en **transaction**, vinculadas con los campos **company_id** con **company**(**id**) y **credit_card_id** con **credit_card**(**id**).
Una vez se añaden los registros con las claves primarias en las tablas correspondientes y se vuelve a probar la inserción, funciona.

- Ejercicio 4

Desde recursos humanos te solicitan eliminar la columna "pan" de la tabla **credit_card**. Recuerda mostrar el cambio realizado.

```
ALTER TABLE credit_card
DROP COLUMN pan;
```

Output

Action Output

#	Time	Action	Message	Duration / Fe
5035	12:49:37	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`transactions`.`trans...	0.031 sec
5036	12:58:41	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec

```
DESCRIBE credit_card;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	varchar(15)	NO	PRI	NULL	
iban	varchar(50)	NO		NULL	
pin	char(4)	NO		NULL	
cvv	char(3)	NO		NULL	
expiring_date	char(8)	NO		NULL	

Result 12

Output

Action Output

#	Time	Action	Message	Duration / Fetch
20	12:10:05	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.109 sec
21	12:10:11	DESCRIBE credit_card	5 row(s) returned	0.000 sec / 0.000

Nivel 2

- Ejercicio 1

Elimina de la tabla **transaction** el registro con ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de datos.

```
DELETE FROM transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5036	12:58:41	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec
5037	13:01:02	DELETE FROM transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	1 row(s) affected	0.015 sec

- Ejercicio 2

La sección de marketing desea tener acceso a información específica para realizar análisis y estrategias efectivas. Se ha solicitado crear una vista que proporcione detalles clave sobre las compañías y sus transacciones. Será necesaria que crees una vista llamada **VistaMarketing** que contenga la siguiente información: Nombre de la compañía. Teléfono de contacto. País de residencia. Media de compra realizado por cada compañía. Presenta la vista creada, ordenando los datos de mayor a menor media de compra.

```
CREATE OR REPLACE VIEW VistaMarketing AS
SELECT c.company_name, c.phone, c.country, t.avg_buy
FROM company AS c
INNER JOIN (
    SELECT company_id, AVG(amount) as avg_buy
    FROM transaction
    GROUP BY company_id
) AS t ON c.id = t.company_id
ORDER BY t.avg_buy DESC;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105123	10:33:15	DELETE FROM transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	1 row(s) affected	0.015 sec
105124	10:34:07	CREATE OR REPLACE VIEW VistaMarketing AS SELECT c.company_name, c.phone, c.country, t....	0 row(s) affected	0.015 sec

```
SELECT * FROM vistamarketing;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	phone	country	avg_buy
Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.911333
Pretium Neque Corp.	07 77 48 55 28	Australia	275.578064
Urna Convallis Associates	06 01 24 77 04	United States	273.566925
At Associates	09 56 61 10 65	New Zealand	272.736985
Metus Vitae Associates	08 25 44 40 66	Australia	270.051876
Aliquet Diam Limited	02 76 61 47 46	United States	269.291145
Nec Luctus LLC	02 14 71 75 73	Norway	268.604837
Neque Tellus Incorporated	04 43 18 34 19	Ireland	267.563349
Cras Consulting	07 50 10 85 63	Belgium	267.382797
Sed LLC	01 63 16 26 52	Belgium	266.607351
Eget Ipsum Ltd	03 67 44 56 72	United States	266.266545
Tortor Nunc Commodo Com...	05 35 92 77 16	United States	266.264375
Amet Institute	06 33 40 21 33	Australia	265.538889
Fusce Corp.	08 14 97 58 85	United States	265.120449
Ut Semper Foundation	01 60 36 33 06	Sweden	264.729774
Orci Adipiscing Limited	03 18 00 77 81	United Kingdom	264.592607

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Output

Action Output

#	Time	Action	Message	Duration / Fetch
5043	13:21:00	CREATE OR REPLACE VIEW VistaMarketing AS SELECT c.company_name, c.phone, c.country, A...	0 row(s) affected	0.015 sec
5044	13:22:24	SELECT * FROM vistamarketing LIMIT 0, 1000	100 row(s) returned	0.547 sec / 0.0

- Ejercicio 3

Filtra la vista **VistaMarketing** para mostrar solo las compañías que tienen su país de residencia "Germany"

```
84 •  SELECT *
85    FROM vistamarketing
86    WHERE country = 'Germany';
87
88
89
90
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	phone	country	avg_buy
Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.911333
Nunc Interdum Incorporated	05 18 15 48 13	Germany	259.319156
Convallis In Incorporated	06 66 57 29 50	Germany	257.693651
Ac Industries	09 34 65 40 60	Germany	255.169777
Rutrum Non Inc.	02 66 31 61 09	Germany	255.137959
Auctor Mauris Corp.	05 62 87 14 41	Germany	254.675099
Augue Foundation	06 88 43 15 63	Germany	253.564644
Aliquam PC	01 45 73 52 16	Germany	252.958601

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Output

Action Output

#	Time	Action	Message	Duration / Fetch
5044	13:22:24	SELECT * FROM vistamarketing LIMIT 0, 1000	100 row(s) returned	0.547 sec / 0.0
5045	13:25:42	SELECT * FROM vistamarketing WHERE country = 'Germany' LIMIT 0, 1000	8 row(s) returned	0.156 sec / 0.0

Nivel 3

- Ejercicio 1

La próxima semana tendrás una nueva reunión con los gerentes de marketing. Un compañero de tu equipo realizó modificaciones en la base de datos, pero no recuerda cómo las realizó. Te pide que lo ayudes a dejar los comandos ejecutados para obtener el

Diagrama esperado:

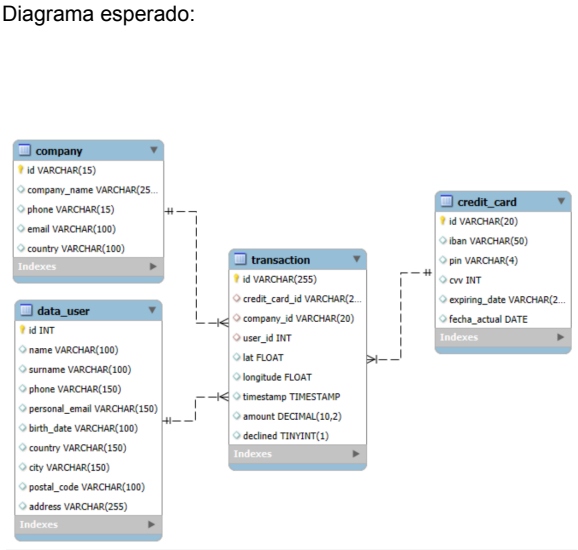


Diagrama actual:

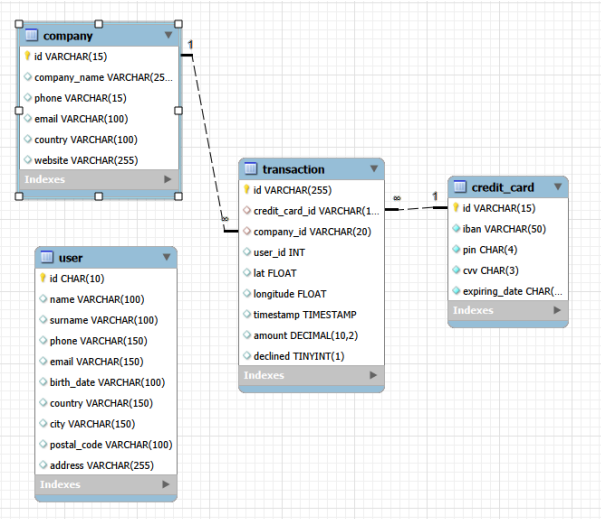
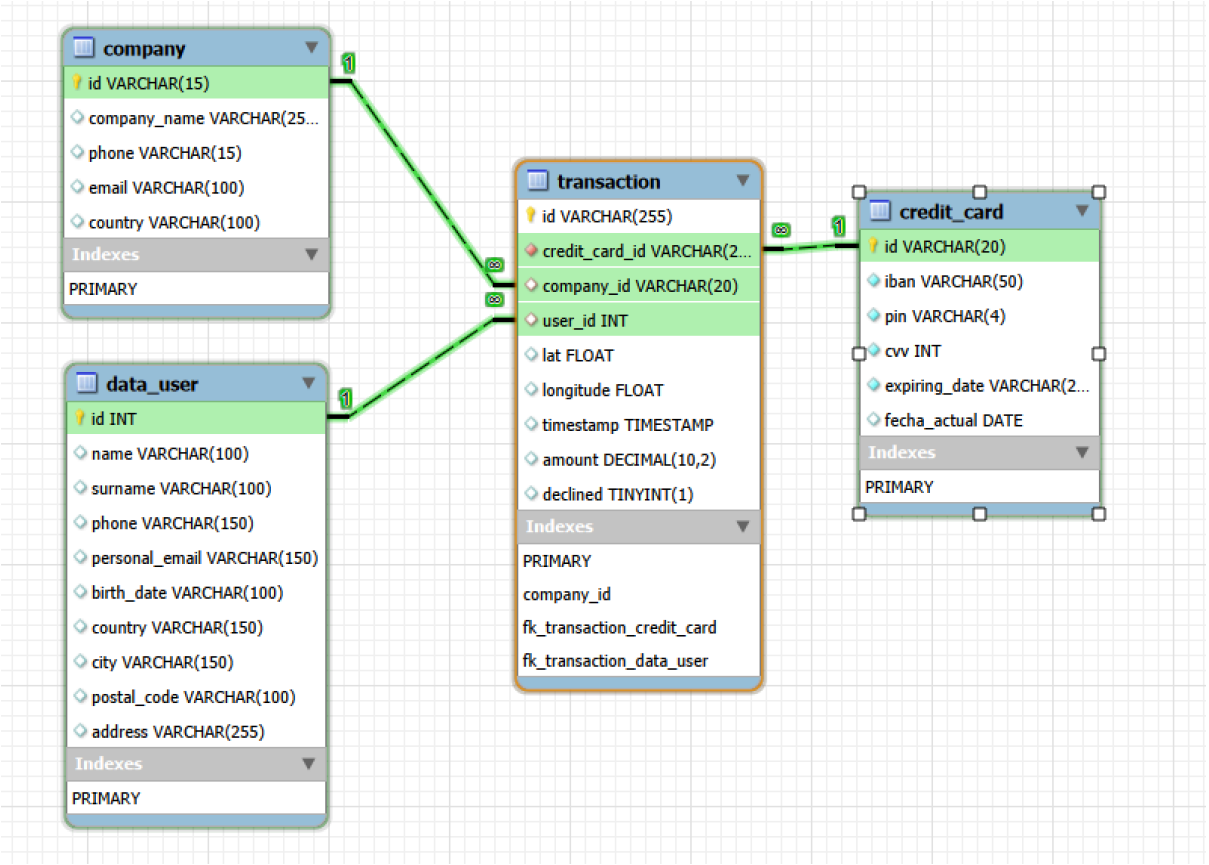


Diagrama resultante:

Aplicado todos los cambios solicitados, se aprecia una incoherencia entre el campo **id** de la tabla **company** VARCHAR(15) y el campo **company_id** VARCHAR(20) de la tabla **transaction**,



Secuencia de instrucciones para las modificaciones:

```
ALTER TABLE company DROP COLUMN website;
ALTER TABLE credit_card
    MODIFY COLUMN id VARCHAR(20) NOT NULL,
    MODIFY COLUMN pin VARCHAR(4) NOT NULL,
    MODIFY COLUMN cvv INT NOT NULL,
    MODIFY COLUMN expiring_date VARCHAR(20) NOT NULL,
    ADD COLUMN fecha_actual DATE DEFAULT NULL;

ALTER TABLE transaction
    MODIFY COLUMN credit_card_id VARCHAR(20) NOT NULL;

RENAME TABLE user TO data_user;
ALTER TABLE data_user
    DROP PRIMARY KEY,
    MODIFY COLUMN id INT PRIMARY KEY,
    RENAME COLUMN email TO personal_email;

ALTER TABLE transaction
ADD CONSTRAINT fk_transaction_data_user
FOREIGN KEY (user_id)
REFERENCES data_user(id);
```

Output

Action Output

#	Time	Action	Message	Duration /
1	11:10:17	ALTER TABLE company DROP COLUMN website	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.078 sec
2	11:10:24	ALTER TABLE credit_card MODIFY COLUMN id VARCHAR(20) NOT NULL, MODIFY COLUMN pin ...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.125 sec
3	11:10:29	ALTER TABLE transaction MODIFY COLUMN credit_card_id VARCHAR(20) NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.172 sec
4	11:10:33	RENAME TABLE user TO data_user	0 row(s) affected	0.094 sec
5	11:10:33	ALTER TABLE data_user DROP PRIMARY KEY, MODIFY COLUMN id INT PRIMARY KEY, RENA...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.344 sec
6	11:10:39	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_data_user FOREIGN KEY (user_id) REF...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.171 sec

- Ejercicio 2

La empresa también os pide crear una vista llamada "**InformeTecnico**" que contenga la siguiente información:

- ID de la transacción
- Nombre del usuario/aria
- Apellido del usuario/aria
- IBAN de la tarjeta de crédito usada.
- Nombre de la compañía de la transacción realizada.

Aseguraos de incluir información relevante de las tablas que conoceréis y utilizar alias para cambiar de nombre columnas según haga falta.
Muestra los resultados de la vista, ordena los resultados de forma descendente en función de la variable ID de transacción.

```
CREATE OR REPLACE VIEW InformeTecnico AS
SELECT
        t.id AS id_transaction,
        du.name AS user_name,
        du.surname AS user_surname,
        cc.iban AS credit_card_iban,
        c.company_name
FROM transaction AS t
INNER JOIN company AS c ON c.id = t.company_id
INNER JOIN credit_card AS cc ON cc.id = t.credit_card_id
INNER JOIN data_user AS du ON du.id = t.user_id
WHERE t.declined = 0
ORDER BY t.id DESC;
```

Output

Action Output

#	Time	Action	Message	Duration / Fe
13	17:04:46	SELECT t.id AS id_transaction, du.name AS user_name, du.surname AS user_surname, cc.iban AS credit_card_iban, c.company_name...	1000 row(s) returned	0.734 sec / (
14	17:07:46	CREATE OR REPLACE VIEW InformeTecnico AS SELECT t.id AS id_transaction, du.name AS user_name, du.surname AS user...	0 row(s) affected	0.031 sec

Listado resultante:

```
SELECT * FROM InformeTecnico;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id_transaction	user_name	user_surname	credit_card_iban	company_name
FFFD31D6-9495-47CE-B54A-7DB8E1CC274B	Bmrgli	Tprvvmrc	XX7948144512112891824909 22	Turpis Company
FFFCF76D-ECF0-4985-A2D0-B2A7B75998FC	Dfrled	Vilqcjdl	XX6362517016478920366760 34	Amet Nulla Donec Corporation
FFFC9E8D-27C7-4ADE-98F2-7533EF4DF126	Securp	Faofvqfy	XX1626771433042236314375 67	Nunc Interdum Incorporated
FFFB270D-F53A-4D5D-9666-E5307C53CC84	Ggzjpa	Uirzjulh	XX3951142670820199525670 52	Viverra Donec Foundation
FFF9E3CE-234E-408C-A8EF-F9CAD577224A	Yshimq	Zpsjsleed	XX8845462156537570367941	Convallis In Incorporated
FFF9E178-6CD2-4DF9-99B0-49AE068809B1	Jevepx	Xwcwzwnm	XX3214055157116543847114 81	Mus Aenean Eget Foundation
FFF867C9-17B5-4B1F-AFD9-F8023AAA449E	Fqlngd	Lvhfqyxi	XX2784463429326809797294 26	Cras Vehicula Aliquet Industries
FFF7042D-18C6-4DDD-823C-4D90A4AC8F26	Njoraa	Egsqcuii	XX4050092725725500820272 09	Placerat LLP
FFF660D4-4244-47F6-9210-E5D1DCB99DB0	Lopzaj	Itgryfay	XX6337665973662745401512 5	Pede Cum Ltd
FFF5C660-4441-436D-BD27-E6C53B618622	Gmnbnu	Oxdvhkll	XX2378202561726463940164 83	At Associates
FFF54F54-B439-41F0-BDD2-F7332DC1ACAD	Gqcfyy	Mpifnltn	XX8027239432401476121587 18	Enim Condimentum Ltd
FFF42F7D-7A0D-46E2-AF72-5996903F8AA9	Ddkugq	Ycsbpryu	XX9264423015551959741995 41	At Pede Corp.
FFF42620-1968-4A1F-B9D7-27843CB097A5	Sjbrvp	Eiqnzdfj	XX3954572326389591023368 73	Amet Nulla Donec Corporation
FFF290EB-79D2-497C-BDCE-2EC63FE3E9AD	Wcfzys	Dkkthgtd	XX4589890482222307175199 35	Integer Mollis Corp.
FFF20FB4-9014-4AC3-B938-E31E7786C49A	Afjbmt	Yshvmtqk	XX1484835527434437357121 34	Ac Fermentum Incorporated

InformeTecnico 8

Output

Action Output

#	Time	Action	Message	Duration / F
110140	11:00:55	SELECT * FROM InformeTecnico LIMIT 0, 1000	1000 row(s) returned	0.000 sec /